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Dear Daniel,

# Syama SAG pebble (scat) ore sorting — assessment and recommendation

Thank you for providing the scat assay data. We have completed the assessment you requested: updating the pebble sorting business case for Syama and testing whether STEINERT ore sorting is worth pursuing on your SAG scat stream. The opportunity is real but conditional, and the next spend should be on test work rather than on plant.

The work builds on the Sukari pebble sorting simulation (ref. SR241558), which rejects barren scat mass, reduces the circulating load and refills the freed SAG capacity with ROM. We re-parameterised it for the converted (ex-oxide) SAG line as it will run after the Sulphide Conversion Project, calibrating to your measured scat stream. The engine transfers unmodified; the material property that made the Sukari case work does not.

## Principal finding

**Your measured scat grade of 2.0 g/t Au is the pivotal number.** Sukari's scats assayed 0.56 g/t against a 1.57 g/t ROM — barren material, discardable almost for free. Syama's run at 83% of ROM grade. Back-solving the deportment against 20% sublevel-cave dilution gives a waste-to-ore contrast of only 1.8×, against 8.9× at Sukari, so roughly one third of the scat stream is barren dilution — a hard ceiling on what any sorter can reject.

*Table 1: Syama scat stream compared with the Sukari base case*

| Parameter | Sukari | Syama |
|---|---|---|
| **ROM / scat grade (g/t Au)** | 1.57 / 0.56 | **2.40 / 2.00** |
| **Scat grade as % of ROM** | 35% | **83%** |
| **Heterogeneity contrast** | 8.9× | **1.8×** |
| **Barren mass available in scats** | ~69% | **~33%** |
| **Reject grade achieved (g/t Au)** | 0.148 | **0.429** |

Every rejected tonne is therefore expensive. At 0.43 g/t and US$4,000/oz the reject stream holds some US$41 per tonne of recoverable gold against roughly US$10 per tonne of cost avoided, so the sorter must be tuned conservatively for high ore recovery — capping mass pull at about 25%.

## Economics

Your scats already recirculate, and the Sulphide Conversion Project is installing a pebble crusher for this stream, so no gold is being lost today. The case is throughput debottlenecking, not recovery, and turns on whether the freed SAG capacity can be refilled with ore.

*Table 2: Outcomes at 95% ore recovery / 70% waste rejection, US$4.1 M capital*

| Scenario | Net US$/a | NPV @ 10% |
|---|---|---|
| **Freed capacity refilled at 2.4 g/t** | +7.58 M | **+33.6 M** |
| **Breakeven (29% capture)** | +0.82 M | **0** |
| **No spare ore — ROM feed held flat** | −1.96 M | **−13.9 M** |

The upside is genuine, with a payback near six and a half months, but the downside is equally real: a plant that cannot fill the capacity it frees destroys value and permanently sends gold to the dump. Price is not the swing factor — at US$2,500/oz the favourable case still returns some US$13 M.

## Recommendation

**We recommend a conditional go: fund the test work, not the plant.** Two gates should close before any capital is committed.

1. **Constraint audit —** confirm that comminution, rather than the roaster or ore supply, binds the converted line, and that ore exists at reserve grade to fill it. The published record points the other way, with sulphide stockpiles drawn down through 2025 to maintain throughput. No cost, about four weeks.
2. **Bulk sensor test —** 2 to 5 tonnes of scats through an XRT rig. A 1.8× mass contrast is not evidence of a detectable sensor contrast, and Syama's gold is refractory, so reject grades need fire assay. Roughly US$60,000 to US$150,000, about three months.

## Related considerations

- We would scope dilution rejection on coarse ROM into the same campaign; 20% dilution across some 2.6 Mt/a of underground feed offers far more heterogeneity than the scats do.
- A sulphidic reject stream will need acid rock drainage characterisation before it can be dumped, carrying its own closure cost.

Absent site data, several model inputs remain placeholders, flagged on the enclosed model's Dashboard; dilution in particular drives the result. We would welcome the chance to refine these with your team.

Yours sincerely,

**Mike Daniel**
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**Enclosures:** decision brief; Syama_Pebble_Sorting_Simulation.xlsx